LUMINARA: SMART STREETLIGHT MANAGEMENT SYSTEM WITH CENTRALIZED SWITCHING AND PER-LAMP FAULT DETECTION FOR BARANGAY 6, CARMONA, CAVITE

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INTRODUCTION

Public street lighting is a basic municipal service that supports road safety, crime prevention, and nighttime mobility (Murray & Feng, 2016). The City of Carmona, Cavite, was classified as a city in 2023 and had a population of 112,140 in 2024 across 14 barangays (Philippine Statistics Authority, 2025). It is a rapidly industrializing urban center with expanding commercial and residential areas, so reliable street lighting along its roads and alleys is essential to the safety of residents.

Despite this importance, the maintenance of streetlights in many local communities remains largely manual and reactive. In Carmona, the Engineering Department is responsible for maintaining public streetlights but keeps no census or maintenance record of them. Most streetlights are switched on and off by hand at a breaker, and repairs are carried out only

after residents report a fault, either by a manual report or through the BML Carmona App. As a result, faults can go undetected for an unknown length of time, and no data exists on lamp operating hours, failure frequency, or repair time. Schneider et al. (2024) observed that patrolling every lamppost at night is impractical, and that relying on citizen reports makes defects hard to locate precisely and produces duplicate reports for the same fault.

Advances in the Internet of Things (IoT) have led researchers to apply sensing and communication technologies to street lighting. Studies have used IoT to conserve energy in street lighting (Yusoff et al., 2020), to centrally monitor streetlights and detect faults using sensors (Vamsi et al., 2024), and to review the wider development of intelligent street lighting for smart cities (Omar et al., 2022). Among communication technologies, LoRa has been used in street lighting because it supports long-range, low-power transmission without requiring network coverage at every pole. Researchers have applied it to smart streetlight systems with fault detection and adaptive illumination (Saranya et al., 2025) and to monitoring retrofitted streetlamps (Schneider et al., 2024).

This is also consistent with the city's vision as a "Smart City" and with Sustainable Development Goal 11, which calls for cities and human settlements that are inclusive, safe, resilient, and sustainable (The Global Goals).

To address these problems, the researchers developed a Smart Streetlight Management System with Centralized Switching and Per-Lamp Fault Detection for Barangay 6, Carmona, Cavite. The system uses an ESP32-based controller with an integrated LoRa transceiver to switch multiple

streetlights from one controller per power feed, with a manual override. It measures total circuit voltage and current using a non-invasive current transformer and a voltage sensing module, and it uses a light sensor to confirm whether a lamp is actually lit. Lamp data is transmitted to a central server, where a web dashboard shows each streetlight's status and fault history.

The developed system gives the Engineering Department and the barangay a verifiable record of streetlight condition, replaces daily manual switching with centralized control, and shortens the time between a fault occurring and being detected. Ultimately, the system is expected to give the barangay and the Engineering Department a dependable basis for maintaining streetlights, and shows how low-cost IoT hardware can support local infrastructure management.

Statement of the Problem

The City of Carmona has no visibility into the condition of its streetlights. The Engineering Department keeps no inventory, maintenance record, or uptime data, and reports that failures are infrequent, but this cannot be verified. Faults are discovered only when residents notice and report them, so unreported outages and the time taken to detect them are unknown. Meanwhile, streetlights in Barangay 6 are switched manually every day, with no record of actual operating hours. The lack of data prevents the city from assessing lighting reliability or planning maintenance.

Specifically, this study aims to answer the following:

1. How can a centralized streetlight switching system with manual override be designed and developed to control the connected streetlight circuit?
2. How accurately can the system identify individual streetlights that appear unlit under expected operating conditions using light-sensor readings?
3. How accurately can the system measure the aggregate electrical parameters of the connected streetlight circuit compared with a suitable reference measuring instrument?
4. How reliably can the system transmit individual lamp-status data and aggregate electrical measurements to the central receiver through LoRa communication?
5. How can a web-based dashboard be developed to display individual streetlight status, aggregate circuit measurements, switching records, and fault indications for monitoring purposes?
6. How does the developed system perform in terms of functional correctness, lamp-status detection accuracy, LoRa communication reliability, electrical measurement accuracy, and switching response during unit, integration, and system testing?
7. How acceptable is the developed system to its intended users in terms of applicable software product quality characteristics, as assessed using ISO/IEC 25010 and

supplemented by applicable quality measures from ISO/IEC 25023?

Objectives of the Study

The main objective of the study is to design, develop, test, and implement a Smart Streetlight Management System that provides centralized switching and per-lamp fault detection, with a web dashboard, for the streetlights of Barangay 6, Carmona, Cavite.

Specifically, the study aims to:

1. Design and develop a centralized streetlight switching system with manual override for controlling the connected streetlight circuit.
2. Implement and evaluate light-sensor-based monitoring to identify individual streetlights that appear unlit under expected operating conditions.
3. Integrate circuit-level electrical monitoring and determine the accuracy of its measurements by comparing the recorded aggregate electrical parameters with a suitable reference measuring instrument.
4. Implement and evaluate LoRa communication for transmitting individual lamp-status data and aggregate electrical measurements to the central receiver.
5. Develop a web-based dashboard that displays individual streetlight status, aggregate circuit measurements, switching records, and fault indications for monitoring purposes.

6. Evaluate the developed system through unit, integration, and system testing in terms of functional correctness, lamp-status detection accuracy, LoRa communication reliability, electrical measurement accuracy, and switching response, and
7. Evaluate the acceptability of the developed system among its intended users using an assessment instrument based on applicable ISO/IEC 25010 software product quality characteristics, supplemented by ISO 25023 quality measurements.

Significance of the Study

The developed system will address the limitations of manual streetlight switching and citizen-dependent fault reporting by providing automated switching, per-lamp fault detection, and centralized records of streetlight status.

The study is expected to benefit the following demographics:

Residents of Barangay 6. Residents will benefit from better-lit streets and alleys, since faulty lamps were detected and identified sooner, improving nighttime safety and pedestrian mobility.

Municipal Office of Carmona's Engineering Department. The department will gain a record of streetlight status, fault history, and uptime, and will be able to identify exactly which lamp had failed. This can then support maintenance planning and more efficient dispatch of repair crews.

Barangay officials and personnel. The barangay will no longer fully rely on personnel and/or citizens to report faulty streetlights, in addition to gaining a digital record log of the lamps' operating hours.

Local government of Carmona. The study supports the city's Smart City vision, in addition to embodying SDG 11, as well as providing data that could support decisions on street lighting investment and maintenance.

Future researchers. This study may serve as a reference for those building IoT-based street lighting systems, particularly ones that use LoRa or energy monitoring.

Time and Place of the Study

The study will be conducted in Barangay 6, Carmona, Cavite, from October 2026 up until the end of the thesis timeline period. System design, development, and laboratory testing will be carried out at Cavite State University – Carmona Campus, while field testing will be carried out on the streetlights of Barangay 6.

Scope and Limitations of the Study

This study focuses on the design, development, testing, and evaluation of LUMINARA, a Smart Streetlight Management System with centralized switching and per-lamp fault detection for selected streetlights within Barangay 6, Carmona, Cavite. The proposed system integrates individual lamp-status monitoring, aggregate circuit-level electrical monitoring, centralized switching, wireless data transmission, and a web-based dashboard. The study will assess the functionality and technical performance of the developed prototype under defined testing conditions. The final number of monitored streetlights will be determined through site inventory, confirmation of the available electrical configuration, and the approved scope of the prototype. Accordingly, the study will not assume that all streetlights

within the barangay are monitored unless the completed system and its testing demonstrate such coverage.

The individual monitoring nodes will use an ESP32-WROOM microcontroller, an SX1278 LoRa transceiver, and a BH1750 illuminance sensor to monitor the apparent illumination of their assigned streetlights. Each node will be associated with a specific monitored lamp and will transmit its sensor readings and available device-status information to the central controller. The proposed communication arrangement will follow a star topology using direct point-to-point LoRa communication and master-initiated polling. Each node will communicate directly with the central controller through an assigned node address and will not forward data for other nodes. Communication procedures will account for response timeouts, missed responses, and retries as applicable to the implemented design. The study will evaluate communication performance under the actual test conditions rather than assume a particular communication range or reliability.

Lamp-status monitoring will primarily rely on illuminance readings interpreted according to the expected switching state and the thresholds established through calibration and testing. The system will indicate whether an assigned streetlight appears lit or unlit under the tested operating conditions. Its ability to distinguish these conditions will be evaluated by comparing the sensor-based classifications with observed lamp conditions. The results may be affected by ambient illumination, light from neighboring luminaires, sensor placement, obstruction, sensor orientation, and other environmental factors. The system will not be assumed to identify every

possible lamp failure or determine its exact cause. In particular, the proposed optical sensing arrangement cannot independently distinguish among a failed LED assembly, a defective driver, disconnected wiring, loss of supply at an individual lamp, or other underlying electrical faults. Detection of flickering or intermittent failures will not be claimed unless the implemented sampling method and corresponding tests demonstrate that the system can identify these conditions reliably.

Circuit-level electrical monitoring will use a PZEM-004T v3 measurement module with a compatible current transformer and the required voltage-sensing connections, subject to verification of the actual module specifications and installation requirements. The module will monitor the aggregate electrical behavior of the connected streetlight circuit. Depending on the implemented module and validated readings, the recorded parameters may include circuit voltage, current, active power, energy consumption, and other supported electrical measurements. The accuracy of the applicable readings will be evaluated through comparison with a suitable reference measuring instrument where feasible. These measurements will represent the monitored circuit as a whole and will not provide separate voltage, current, or power readings for individual streetlights. Aggregate electrical measurements may indicate changes in circuit operation which may assist in general fault detection, but they will not independently establish which lamp caused a change or conclusively identify the underlying fault.

The centralized switching function will control the connected streetlight circuit as a whole through an appropriately rated switching device operated by

the central controller. The design will incorporate a manual override arrangement appropriate to the approved prototype configuration. The study will examine switching functionality, command execution, response time, and relevant operational behavior under defined test conditions. The system will not provide independent switching or dimming of individual streetlights. Any connection to the existing public streetlight installation will require formal authorization, appropriate electrical protection, and installation or inspection by qualified personnel. If authorization or suitable access to the municipal electrical circuit is unavailable, switching and electrical integration tests will be conducted using an approved laboratory or prototype circuit. In that case, the findings will be limited to the tested prototype arrangement and will not be presented as evidence of successful switching or operation of the actual public streetlights.

The central controller will receive the readings and status information transmitted by the individual nodes, acquire the available circuit-level electrical measurements, and provide the information required by the monitoring application. The system will use a wireless connection between the central controller and the selected server or web application infrastructure, subject to the availability and finalization of the communication and software configuration. A web-based dashboard will be developed to display the status of monitored streetlights, available aggregate electrical measurements, switching records, and fault indications. Where supported by the implementation, the system will also record node responses, missed responses, and other relevant communication events. The dashboard is intended to support the review of monitoring information and recorded events;

it will not independently verify physical lamp conditions, diagnose the exact cause of a fault, dispatch repair personnel, or guarantee that maintenance actions will be completed.

The developed prototype will undergo unit, integration, and system testing. Technical evaluation will cover the functionality of the implemented monitoring and switching components, lamp-status detection performance, LoRa communication reliability, the accuracy of applicable circuit-level electrical measurements, and the response of the integrated system to defined commands and test conditions. Lamp-status detection will be assessed against observed conditions using appropriate classification measures, such as accuracy, precision, recall, and false-positive and false-negative rates, where the test data permit their calculation. Communication evaluation may include packet delivery ratio, successful poll responses, missed responses, retries, received signal strength indicator (RSSI), signal-to-noise ratio (SNR), and response time, as supported by the implementation. Electrical measurement error will be determined through comparison with suitable reference readings where feasible. Switching evaluation will include command success, response time, and functional correctness under the tested configuration. The test procedures, sample conditions, measurement methods, and acceptance criteria will be defined in the methodology before the corresponding results are interpreted.

User evaluation will assess the perceived acceptability of the developed system among its identified intended users using an assessment instrument based on applicable software product quality characteristics from

ISO/IEC 25010. The selected characteristics, questionnaire items, respondent selection, and evaluation procedures will be specified in the methodology. This assessment will concern users' perceptions of the system and will be reported separately from objective engineering test results. Favorable user evaluations will not, by themselves, establish technical accuracy, communication reliability, electrical safety, or readiness for municipal deployment. The study will not claim full compliance with ISO/IEC 25010 solely on the basis of a questionnaire.

The individual monitoring nodes are planned to use rechargeable lithium-ion batteries, with a regulated supply for the electronic components and battery-voltage monitoring where implemented. Battery operation introduces constraints related to power consumption, available capacity, charging or replacement, and continued node availability. The study will document relevant power-consumption measurements or calculations where feasible. Battery endurance and a recommended replacement interval will not be claimed as established capabilities unless supported by documented calculations and representative testing. Long-term battery degradation and maintenance requirements over an extended operational period are outside the scope of the study unless explicitly included in the final test plan.

The study is limited to the selected streetlights, the verified site conditions, and the electrical configuration represented in the approved prototype and testing arrangement. Results obtained from the selected site or a laboratory prototype may not generalize to streetlights with different luminaires, electrical configurations, installation conditions, or environmental

characteristics. LoRa performance may vary with distance, obstructions, antenna placement, interference, and the presence of metal poles or enclosures. Lamp-status detection may vary with illumination conditions and sensor installation. Electrical measurement results will be limited to the parameters supported by the installed measurement module and the conditions under which its readings are validated. The study will not establish long-term performance, reliability, or maintenance effectiveness from short-duration functional tests.

The system will not include individual lamp switching, adaptive dimming, mesh networking, LoRaWAN infrastructure, automatic fault-cause diagnosis, predictive maintenance, automated repair-crew dispatch, or integration with the BML Carmona App. It will not replace existing resident reporting channels or the responsible municipal department's maintenance and repair procedures. The study will not establish a complete streetlight inventory for Carmona, cover all barangays, demonstrate citywide deployment, or prepare an implementation plan for municipal adoption. Actual adoption, procurement, installation at municipal scale, and long-term operational management are outside the scope of the study. Any conclusions regarding the system's suitability for broader deployment will therefore be limited to the evidence obtained from the developed prototype and the defined evaluation procedures.

Definition of Terms

To better understand the study, the following terms are defined conceptually and/or operationally:

Aggregate electrical monitoring is the measurement of electrical parameters for the connected streetlight circuit as a whole to observe its operating condition and identify abnormal changes in circuit-level measurements.

Centralized switching is the control of the on and off state of all streetlights from one controller per power feed, instead of manual switching at the breaker.

Current transformer (CT) sensor is a non-invasive sensor that measures the electric current flowing through a lamp's wire without cutting the wire.

Fault indication is a system-generated status that identifies an observed abnormal condition, such as a lamp appearing unlit during an expected operating period or an abnormal change in aggregate circuit measurements.

ISO/IEC 25010 is the international standard defining software product quality characteristics, used in this study as the evaluation instrument.

LoRa (Long Range) communication is a low-power, long-range wireless communication technology used in this study to transmit lamp data from the controller to the receiver unit.

Manual override is a physical means of switching the streetlights on or off regardless of the controller's command.

Per-lamp fault detection is the identification of an individual streetlight that appears unlit while it is expected to operate, based primarily on its assigned light sensor and the system’s established status-detection criteria.

This does not necessarily determine the internal electrical cause of the abnormal condition.

Web dashboard is the interface used to view individual lamp status, circuit-level electrical readings, switching records, and detected abnormalities.

Conceptual Framework of the Study

The Input–Process–Output (IPO) Model shown in Figure 1 illustrates the conceptual framework of the study by presenting its three major phases: Input, Process, and Output.

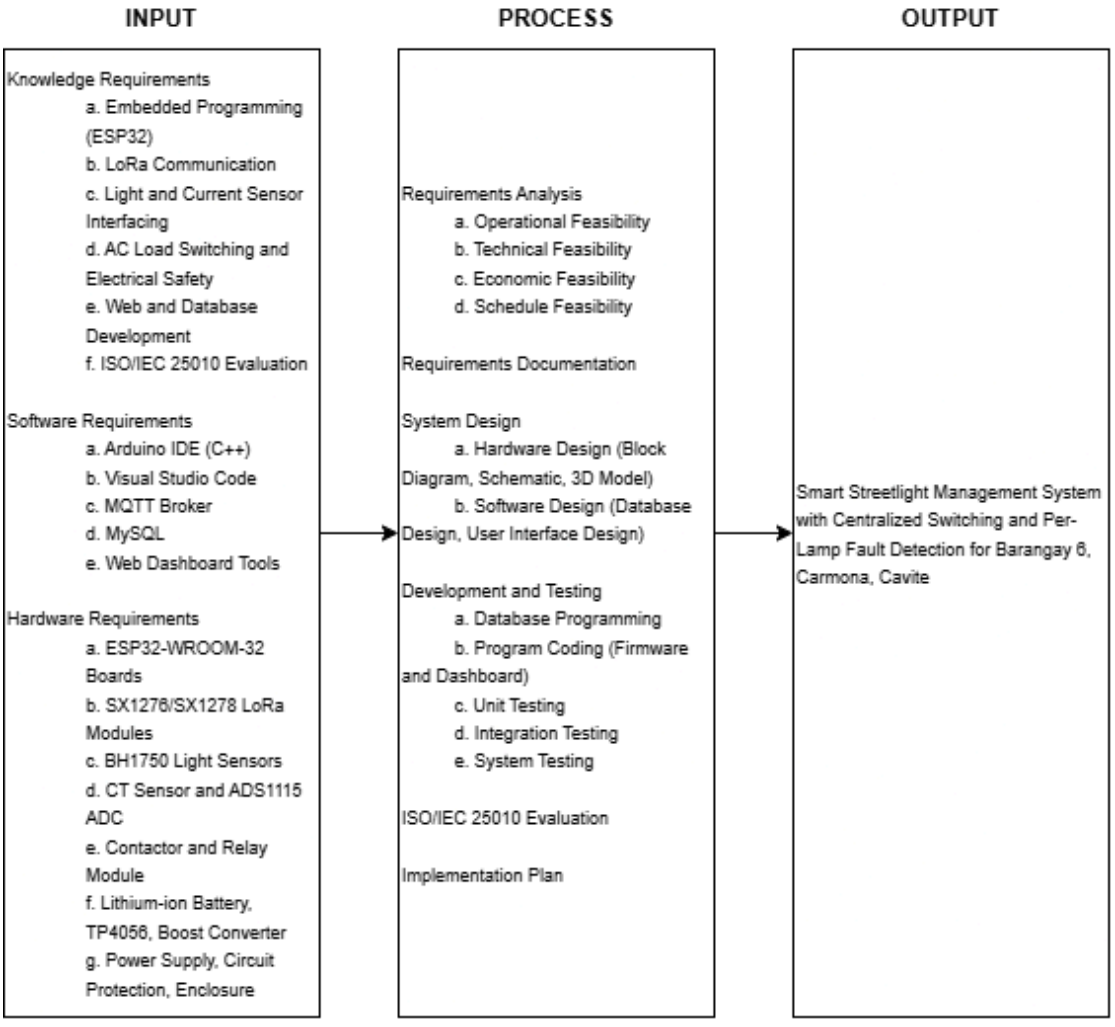


Figure 1. Conceptual Framework of the Study

The Input phase consists of the knowledge, software, and hardware requirements needed to build the system. The knowledge requirements cover embedded programming on the ESP32, LoRa communication, interfacing of the light and current sensors, safe switching of AC loads, web and database development, and the ISO/IEC 25010 evaluation. The software requirements are the Arduino IDE, Visual Studio Code, an MQTT broker, MySQL, and the tools for the web dashboard. The hardware requirements include the ESP32 boards, LoRa modules, light sensors, current sensor, contactor, battery power components, enclosures, and the mockup streetlights.

The Process phase begins with the Requirements Analysis, wherein the operational, technical, economic, and schedule feasibility of the system are examined through interviews and a site survey in Barangay 6. The findings are then recorded in the Requirements Documentation, which serves as the basis of the System Design: the hardware design (functional block diagram, schematic diagrams, and 3D models) and the software design (database and user interface). In the Development and Testing stage, the prototype is built in increments, with the database programmed and the firmware and dashboard coded, and each increment is tested through unit, integration, and system testing. The completed prototype is then evaluated using the ISO/IEC 25010 instrument, and an Implementation Plan is prepared for the potential pilot adoption of the system by the Carmona Engineering Department.

The Output is the Smart Streetlight Management System with Centralized Switching and Per-Lamp Fault Detection for Barangay 6, Carmona, Cavite, a prototype that switches the streetlights from one controller and identifies the exact lamp that has failed, with a web dashboard for monitoring and fault history.

REVIEW OF RELATED LITERATURE

This chapter presents a review of existing literature, technological frameworks, and empirical studies relevant to the design and implementation of an IoT- and LoRa-based smart street lighting monitoring and control system. The literature reviewed in this section establishes the theoretical, technical, and practical foundations of automated fault detection, long-range wireless communication, centralized feeder control, and web-based infrastructure management. Furthermore, this chapter highlights current global and local developments while identifying the specific technological gaps that the proposed system aims to address.

Manual Streetlight Monitoring and Maintenance Practices

Street lighting serves as a critical component of urban infrastructure, directly impacting public safety, crime mitigation, and overall road hazard reduction. According to Schneider et al. (2024), reliable street lighting minimizes traffic accidents, deters criminal activity, and substantially enhances the perceived safety of public spaces. Conversely, unaddressed malfunctions and delayed maintenance severely diminish these security benefits. Lartsey et al. (2021) noted that poorly managed streetlight networks exacerbate road hazard risks, lead to unchecked energy and financial losses, and result in widespread public dissatisfaction. When faults occur, traditional manual inspection routines frequently cause prolonged outages, leaving hazardous roadways unlit for extended periods.

Fault detection in these installations still depends largely on manual inspection and citizen reports. Schneider et al. (2024) explained that patrolling every lamppost, especially at night, is impractical. They added that citizen reports are unreliable because defects are hard to locate precisely without inspection by dedicated personnel, and the same defect is often reported several times. In the same way, Vamsi et al. (2024) described traditional methods as dependent on citizen reports or manual inspections. They proposed automatic IoT-based detection to eliminate manual searching and speed up repairs. Marathe et al. (2025) likewise observed that urban streetlight systems often rely on manual monitoring. In their account, the lack of real-time diagnostics delays fault notice and extends downtime.

To eliminate delays in maintenance, automated monitoring frameworks have emerged as a necessary intervention for rapid hazard response. According to Vamsi et al. (2024), incorporating sensor-enabled monitoring networks allows municipal operators to transition from passive complaint-based reporting to proactive, centralized fault detection. By identifying burnt-out lamps, power failures, or hardware breakdowns instantaneously, these systems facilitate immediate repairs, thereby maintaining consistent road visibility and preventing accidents. Furthermore, Sathisha et al. (2025) and Rajeshwaran et al. (2022) highlighted that integrating real-time location tracking into fault-detection platforms enables maintenance teams to pin down precise fault locations instantly, drastically reducing response delays and optimizing operational workflows.

These studies show that manual switching and report-based maintenance lead to delayed detection, imprecise fault location, and wasted

energy. This supports the need for a system that switches streetlights centrally and identifies the exact faulty lamp, as proposed for Barangay 6.

IoT-Based Street Lighting Systems

Internet of Things (IoT) technology has been widely integrated into urban lighting infrastructure, primarily driven by the goal of optimizing energy efficiency and automating control routines. According to Yusoff et al. (2020), implementing an Arduino-based controller paired with infrared motion sensors in place of traditional timer-based mechanisms successfully reduced streetlamp energy consumption by 45.48% on weekdays and 32.22% on weekends. The system also displayed real-time lamp status on a Blynk interface for maintenance tracking.

Similarly, Chowdhury et al. (2022) proposed a cloud-integrated framework for major urban centers that consolidated ambient light dimming, traffic-adaptive control, and fault detection. Their study reported energy cost reductions of approximately 60%, demonstrating how interconnecting streetlamps with cloud services optimizes operational efficiency.

Beyond energy conservation, IoT architectures enable continuous, remote supervision of hardware conditions. In their review, Omar et al. (2022) noted that modern smart street lighting platforms leverage interconnected sensor arrays for remote diagnostic evaluations of lamp failure. They compared various system limitations, emphasizing the role of IoT in replacing manual inspection routines.

However, implementing IoT systems in municipal settings presents notable technical hurdles. Khemakhem and Krichen (2024) reported that

integration remains challenging due to the management of diverse sensors and actuators, long-term communication reliability, secure data handling, and overall system scaling.

These studies show that IoT is well established for energy-saving control, while fault detection and maintenance are often secondary features. This supports a system that treats lamp monitoring and fault records as core functions and is designed to scale from a small deployment.

LoRa-Based Communication for Infrastructure Monitoring

Long Range (LoRa) technology has emerged as an optimal low-power, long-range wireless communication protocol for streetlamps spread over wide municipal areas. According to Sánchez-Sutil and Cano-Ortega (2020), deploying Arduino-based sensing nodes to measure electrical variables and ambient lighting levels allows efficient real-time telemetry transmission to cloud platforms over a LoRa network.

Similarly, Jabbar et al. (2025) constructed a LoRaWAN-based control architecture utilizing passive infrared (PIR), light-dependent resistor (LDR), and GPS sensors. Their system automated switching and brightness adjustments, effectively reducing the need for manual on-site operation.

In off-grid applications, Vadi (2025) demonstrated that LoRaWAN provides an internet-independent communication backbone. Their study highlighted how LoRaWAN enables reliable remote monitoring and control for solar-battery streetlamp systems without relying on public cellular networks.

LoRa technology has also been applied directly to automated fault detection. Saranya et al. (2025) proposed a LoRa-enabled system

incorporating weather-based adaptive illumination alongside real-time web control for fault monitoring.

In particular, Schneider et al. (2024) validated a retrofit LoRaWAN monitoring framework equipped with light sensors. Their nine-month trial on 52 lampposts successfully detected lamp states without requiring internal modification to the lamp, though radio coverage was affected by structural obstructions. By contrast, the GSM-based system of Sajonia et al. (2025) can strictly be deployed in areas with established cellular coverage.

These findings confirm that LoRa removes dependency on cellular networks and fits scattered feeder points. This provides empirical support for the deployment of LoRa between field controllers and the central gateway in Barangay 6, with coverage tested on site.

Fault Detection Methods

Automated streetlight fault detection generally relies on optical or electrical sensing modalities. Schneider et al. (2024) advocated for optical measurement because it operates independently of lamp type and captures the progressive decline of light output over time. Likewise, Dongardive et al. (2024) monitored light intensity using microcontrollers and cloud databases to detect outages automatically.

Electrical parameter sensing, on the other hand, identifies the root cause of a failure more directly. Lavric et al. (2014) demonstrated the utility of current sensors in identifying electrical malfunctions during field tests of large-scale wireless sensor networks. Similarly, Sajonia et al. (2025) derived specific error condition codes by analyzing current and voltage

cross-readings. Their system identified distinct faults, such as zero bulb current despite applied voltage at night, or active current draw during daytime hours.

To further improve accuracy, Vamsi et al. (2024) combined LDR light sensors with ACS712 current sensors to gather concurrent data on light output and wiring integrity. By applying a machine learning model, their system predicted lamp status with 98.8% accuracy.

Because optical-only detection reveals that a lamp is dark but cannot explain why, combining current, voltage, and light sensing offers a significant advantage. As implemented in the proposed system, multi-sensor verification allows the system to confirm a failure and distinguish a line-level feed outage from an isolated lamp defect.

Centralized Switching and Control Systems

Centralized control architectures replace manual, per-pole operation with remote line switching. Early field tests by Lavric et al. (2014) and Sánchez-Sutil and Cano-Ortega (2020) validated the execution of remote switching commands across distributed wireless sensor networks.

Building on centralized control, Jabbar et al. (2025) automated switching routines to eliminate manual intervention. Furthermore, Saranya et al. (2025) integrated real-time web-based control, while Sajonia et al. (2025) enabled remote operation via web and mobile dashboards, drastically reducing the labor required for physical inspections.

Control at the line level is also well documented for municipal infrastructure. Schneider et al. (2024) described an electrical monitoring

system operating on entire lamp lines through control nodes installed at line switches. This configuration operates directly at the feeder level.

Switching at the feeder level fits the existing breaker-controlled lines of Barangay 6 and avoids replacing every individual lamp fixture. This provides strong justification for centralized switching paired with a manual override mechanism for safety.

Web-Based Dashboards for Municipal Infrastructure Monitoring

Web-based dashboards serve as the core interface that translates raw sensor telemetry into actionable information for municipal maintenance teams. Daely et al. (2017) designed a centralized web server receiving real-time data from LED streetlamps, giving authorized operators a flexible web interface for status tracking.

Similarly, Vamsi et al. (2024) displayed predicted status and automated fault details on a central web page. This interface eliminated manual searches and accelerated repair workflows. Dongardive et al. (2024) also utilized a Flask-based web dashboard to visualize real-time lamp conditions and issue automated fault alerts.

In a localized implementation, Sajonia et al. (2025) built web and mobile interfaces featuring account authentication and map-based navigation from province to barangay level. Each barangay listed its individual streetlights, and error codes appeared on the dashboard during active malfunctions.

These implementations demonstrate that a web dashboard serves effectively as both a live monitoring tool and a permanent maintenance ledger.

This reinforces the need for a central dashboard that provides the Carmona Engineering Department with a comprehensive lamp inventory and historical fault records, which it currently lacks.

Local Studies (Philippines) and SDG

Local literature on automated street light monitoring in the Philippines remains limited. Sajonia et al. (2025) of Caraga State University addressed high maintenance costs and the lack of real-time monitoring in Geographically Isolated and Disadvantaged Areas (GIDA). Their ESP32 and GSM system monitors solar-battery parameters through a dashboard, supporting the national mandate for total electrification by 2028. However, its scope is restricted to solar-powered luminaires and regions with GSM connectivity.

On a commercial scale, media reports highlight Puerto Princesa as the first Philippine city with connected LED street lighting. Industry representatives noted that the commercial system measures electricity consumption and assists city managers in planning maintenance schedules. Additionally, joint assessments by the World Bank and the Department of the Interior and Local Government (DILG, 2024) recommend that local government units adopt smart city frameworks to improve service delivery.

The development of automated street lighting also aligns with United Nations Sustainable Development Goals: SDG 11 (Sustainable Cities and Communities) (United Nations, 2015). Schneider et al. (2024) directly linked street lighting to accident prevention, lower crime rates, and enhanced public safety, directly supporting SDG 11.

Synthesis

The reviewed literature reveals three primary trends in smart street lighting technology. Most existing research focuses heavily on energy savings through automated dimming and motion-based control mechanisms. In terms of diagnostic capabilities, fault detection is predominantly built on single-variable sensing—either purely optical or purely electrical—with limited studies combining both for comprehensive failure diagnosis. Furthermore, LoRa networks and web-based dashboards have established themselves as standard choices for long-range communication and centralized data visualization in wide-area infrastructure monitoring.

Global studies emphasize purpose-built or completely new smart lighting installations, whereas local Philippine developments remain sparse. Philippine literature consists primarily of off-grid, GSM-dependent solar platforms designed for remote communities or expensive commercial citywide rollouts in major urban centers. In practice, municipalities often retain legacy lamps due to the prohibitive cost of total infrastructure replacement, making retrofit solutions essential.

A clear technological and contextual gap exists within the reviewed literature. None of the existing studies combine centralized feeder-level switching for manually operated circuits, per-lamp fault verification using concurrent current, voltage, and light sensing, and a centralized web dashboard that builds an automated maintenance record for a grid-powered, barangay-level network.

The proposed system directly addresses this gap. By deploying an ESP32-LoRa controller with manual override at the feeder level, multi-sensor status verification per lamp, and a web dashboard to log lamp and fault records, the study provides a practical, low-cost solution for municipal adoption.

METHODOLOGY

This chapter outlines the research design, hardware and software materials, system architecture, development processes, testing procedures, evaluation instruments, data analysis methods, objective validation matrix, and implementation strategy used in the design, development, testing, and evaluation of the Smart Streetlight Management System with Centralized Switching and Per-Lamp Fault Detection for Barangay 6, Carmona, Cavite.

Research and Development Design

This study employs an engineering research and development (R&D) design focused on prototyping, system integration, and experimental field testing. The research design combines hardware prototyping, embedded firmware execution, web server application development, and telecommunication network integration via Long Range (LoRa) technology. The methodology supports the systematic fulfillment of the project's specific objectives: designing the centralized switching and sensing architecture, developing the embedded nodes and server backend, evaluating system acceptability using the ISO/IEC 25010 software quality standard, and formulating a pilot implementation plan for the Carmona Engineering Department.

Materials

Materials needed for the prototype. The prototype is made up of two kinds of hardware units: the *lamp node*, a small battery-powered unit placed beside each streetlight to check whether its lamp is lit, and the *panel*

controller, a single unit installed at the power feed that switches the lamps on and off, measures the feed's current, and receives the readings of every lamp node. The following materials provide the hardware needed for sensing, wireless communication, switching, and power. Items shared by both units come first, followed by those exclusive to the lamp node and to the panel controller.

The development of the Smart Streetlight Management System requires a combination of hardware components, power electronics, sensing devices, communication hardware, server infrastructure, and software tools.

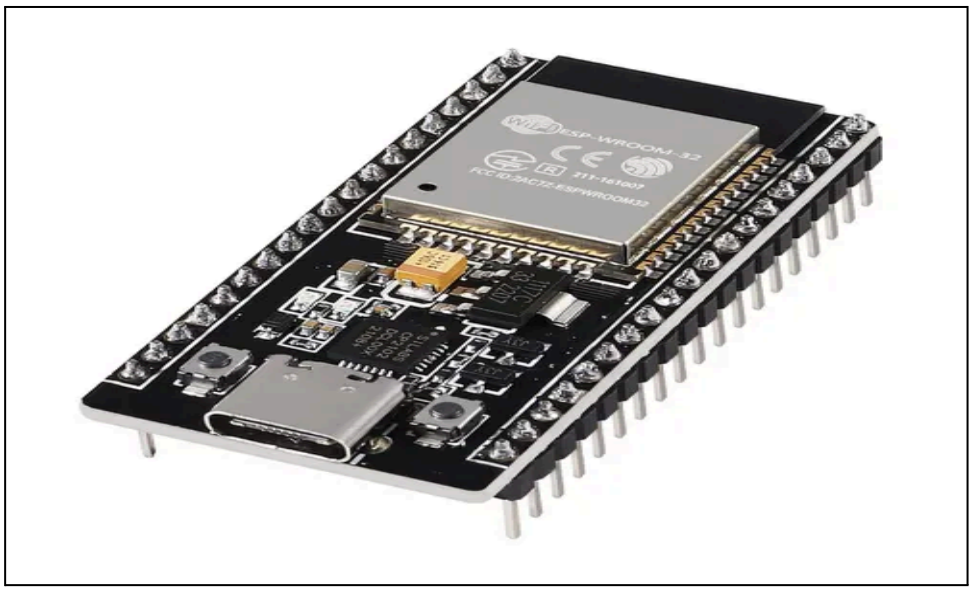


Figure 2. ESP-32 WROOM-32

ESP32-WROOM-32 development board. The ESP32-WROOM-32 development board featured in Figure 2 serves as the controller of both hardware units. Its dual-core processor and built-in Wi-Fi (Espressif Systems,

n.d.) allow the same board to play two roles. In the lamp node, it reads the light sensor and answers the panel controller's requests. In the panel controller, it polls the lamp nodes, reads the current sensor, drives the contactor, and forwards the collected data to the server over Wi-Fi. Using one board for both roles keeps the cost low and lets the researchers reuse the same firmware libraries on both ends.



Figure 3. SX1278 Ra02 LoRa Module

SX126/SX1278 LoRa Module. The SX1276/SX1278 LoRa module in Figure 3 serves as the wireless link between the lamp nodes and the panel controller. LoRa (Long Range) is a low-power, long-range radio technology, hence a lamp node can reach the panel controller without needing Wi-Fi or cellular coverage at every pole (Semtech, n.d.). The modules operate as plain point-to-point LoRa (LoRaWAN is not used), where the panel controller's

module acts as the single receiver for all lamp nodes. Each module talks to its ESP32 through SPI.

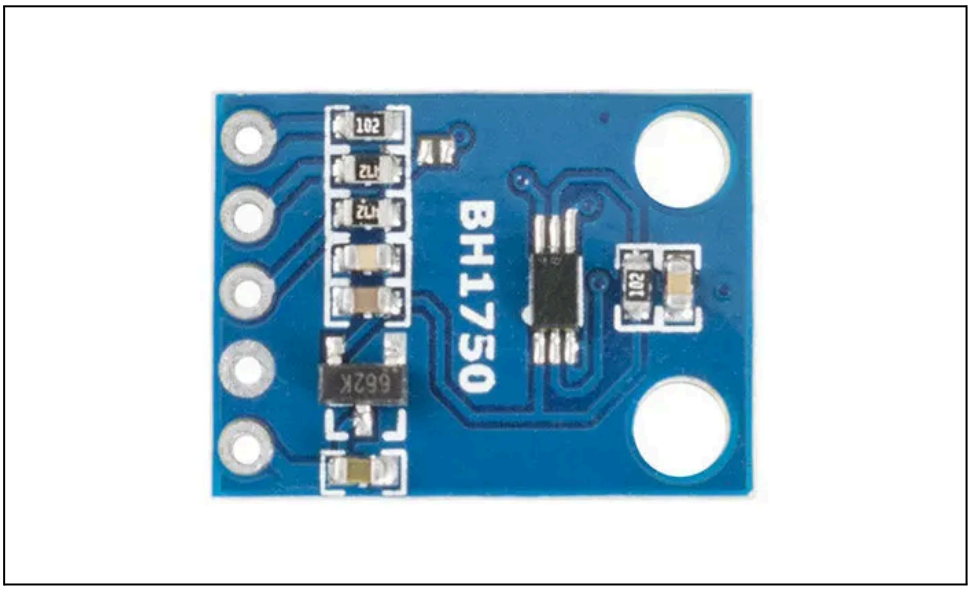


Figure 4. BH1750 Sensor

BH1750 Ambient Light Sensor. The BH1750 module shown in Figure 4 acts as the lamp node's "eye." It is a digital sensor that reports light intensity in lux over I2C, mounted inside the lamp node and aimed at the bulb to determine whether the lamp is actually emitting light. Since it returns a numerical value rather than a mere on/off signal, the threshold for what counts as "lit" can be tuned during calibration, and flickering can be recognized from the variation between consecutive readings.



Figure 5. Lithium-ion Battery

Lithium-ion battery. The lithium-ion battery in Figure 5 powers the lamp node. A battery is used on purpose: the node does not need to be wired to the lamp's mains line and, more importantly, it stays alive when the lamp's power is cut. A node powered by the lamp's own line would go silent during a power failure, making a dead lamp indistinguishable from a dead node.

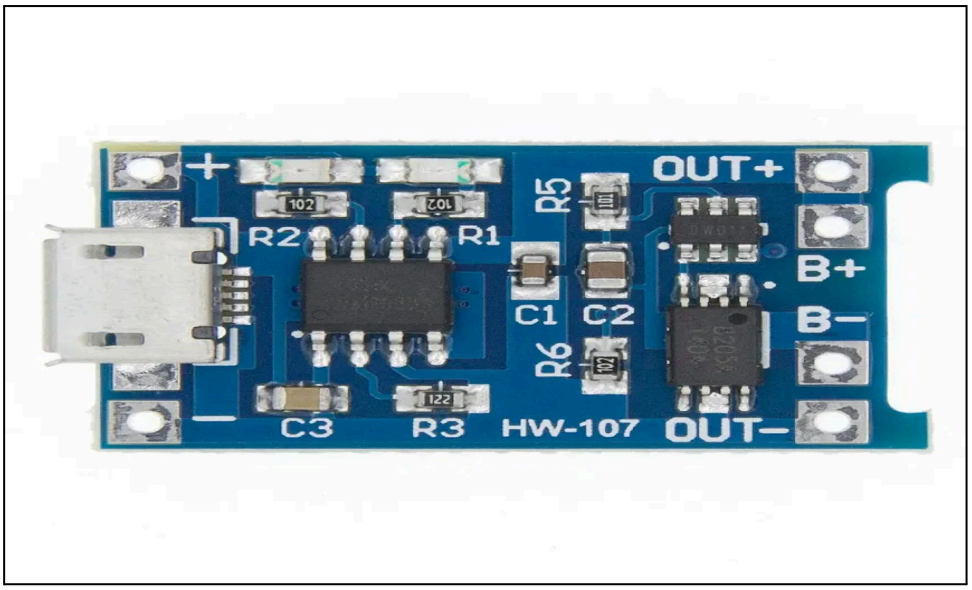


Figure 6. TP4056 Charging Module

TP4056 charging module. The TP4056 module in Figure 6 charges the lithium-ion battery from a standard USB cable and stops once the cell is full. This allows the battery to be topped up (or swapped) during maintenance rounds without any special charger.

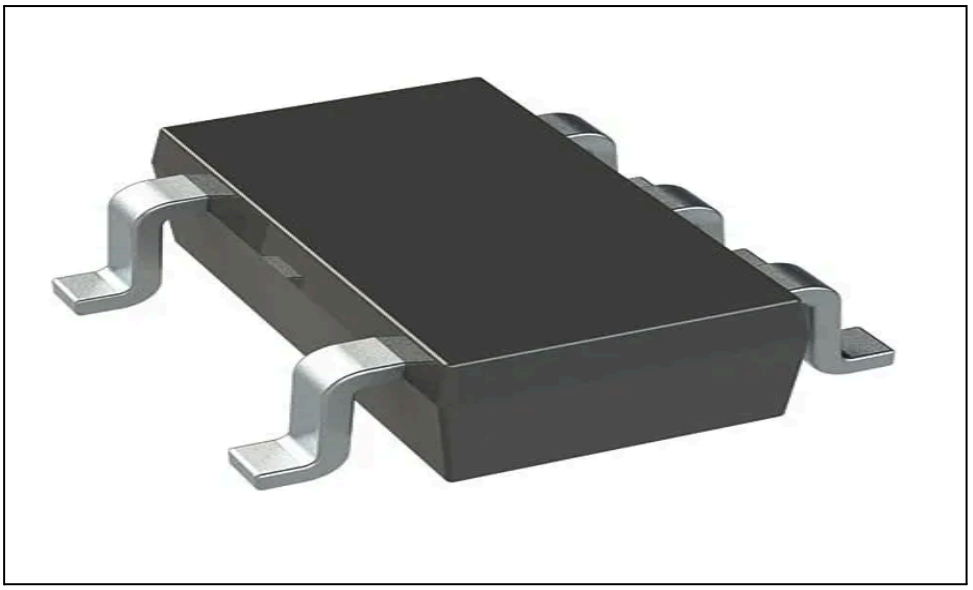


Figure 7. AP2112K-3.3TRG1 IC REG LDO 3.3V

AP2112K LDO (Low Dropout) Linear Voltage Regulator. Figure 7 delivers a fixed output voltage (commonly 3.3V, but also available in 1.2V, 1.8V, 2.5V, etc.) from an input range of 0.8V to 6V, with a typical dropout voltage of only 250mV at full load. Its low quiescent current (~55µA) and standby current (0.01µA) make it ideal for battery-powered devices like portable DVD players, laptops, and IoT sensors, helping to extend battery life. It includes built-in over-current, thermal shutdown, and short-circuit protection features to ensure reliable operation in microcontroller circuits, LCD monitors, and embedded systems.

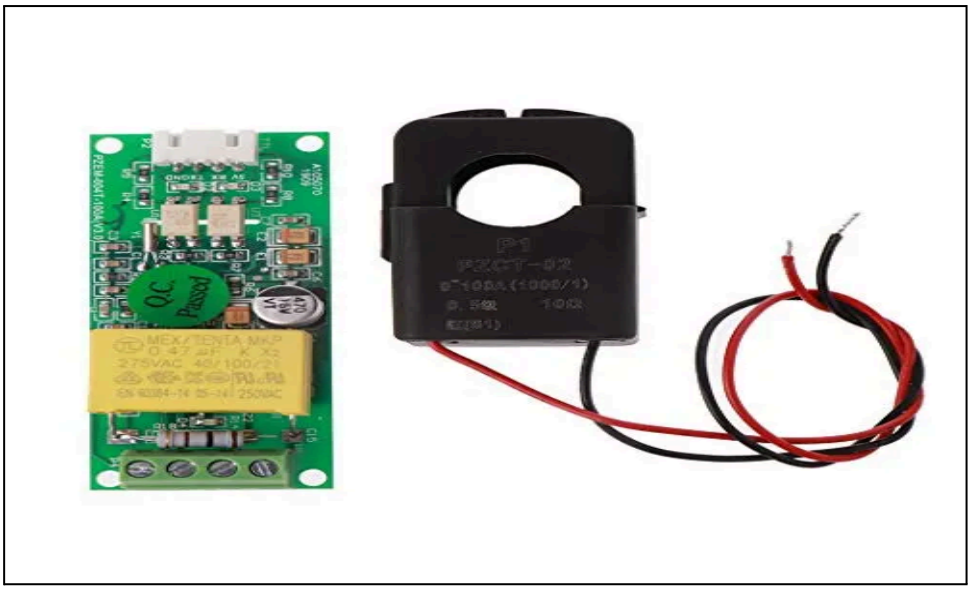


Figure 8. PZEM-004T Non-Invasive CT Sensor

Non-invasive CT sensor. The current transformer (CT) sensor in Figure 8 measures the alternating current flowing through the feed's live wire by clamping around it, without cutting or stripping the conductor. Since the lamps on a feed are wired in parallel, one sensor reads the combined current of every lamp on that feed. Each lamp that goes dark lowers this reading by roughly its own rating (about 0.45 A for a 100 W lamp at 220 V), hence the CT provides electrical evidence that supports the light sensors' findings.



Figure 9. Chint NCH8-25/20

Chint NCH8 Contactor. The contactor in Figure 9 switches the streetlight feed on and off. A contactor is an electrically operated switch built for the heavier AC loads of a streetlight line, and since all lamps on the feed share one line, a single contactor switches every lamp at once. The relay module beside it serves as the “middleman”: the ESP32’s pin can only provide a tiny signal, which the relay module uses to energize the contactor’s coil whilst keeping the low-voltage electronics isolated from the mains.

Circuit Protection Components. Includes fuses, surge protection devices, and circuit breakers to safeguard hardware from transient overvoltages and electrical faults.

Outdoor Weatherproof Enclosure. Houses electronics, sensing modules, and relays in a sealed, weather-resistant enclosure rated for outdoor deployment.

Material Costs

Table 1 summarizes the required materials, quantities, unit prices, and overall project development cost for the Smart Streetlight Management System.

MATERIAL / COMPONENT	QUANTITY	PRICE PER UNIT (PHP)	TOTAL PRICE (PHP)
ESP32-WROOM-32 development board	5	₱350.00	₱1,750.00
SX1278 LoRa Transceiver Module	5	₱450.00	₱2,250.00
BH1750 ambient light sensor module	5	₱80.00	₱400.00
Lithium-ion battery (with holder)	5	₱200.00	₱1,000.00
TP4056 lithium battery charging module	5	₱40.00	₱200.00
AP2112K LDO	1 set (10 pcs)	₱5.00	₱50.00
PZEM-004T v3.0 100 A with split-core CT	1	₱380.00	₱380.00
Chint NCH8-25/20, 2P, 230 V coil Contactor	1	₱720.00	₱720.00
5V Relay Module	1	₱120.00	₱120.00
Manual Override Switch	1	₱150.00	₱150.00
AC-DC Power Supply Module (5V, 1A)	1	₱350.00	₱350.00
Circuit protection set, miscellaneous (Fuse 2A)	1	₱400.00	₱400.00
TOTAL ESTIMATED PROJECT COST			₱7,770.00

Table 1. Material cost computation

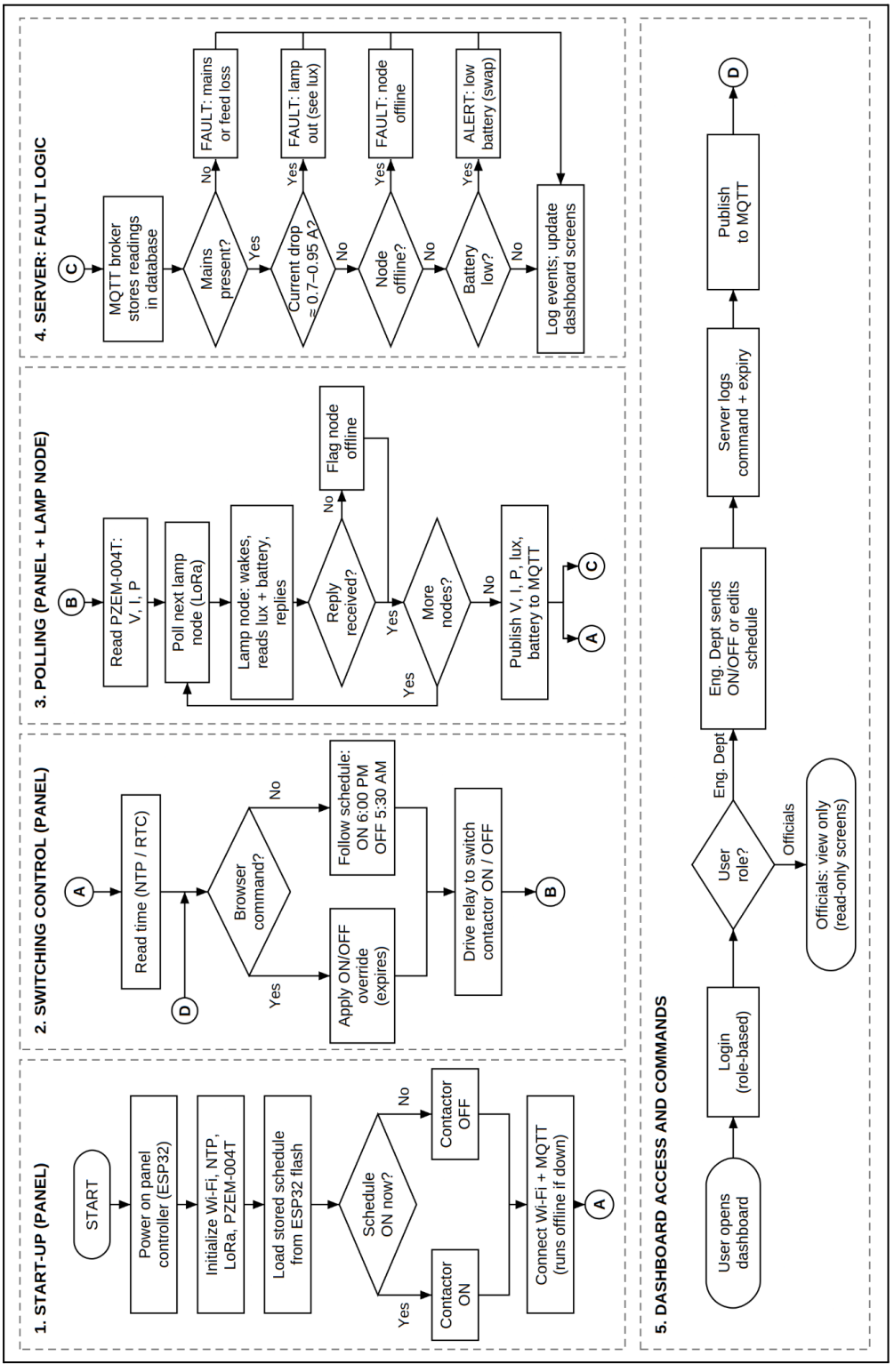


Figure 10. General Process Flow Diagram

General process flow diagram. The general process flow diagram in Figure 10 illustrates the operational sequence of the system, from the start-up of the panel controller up to the display of results and the sending of commands on the web dashboard. It is divided into five parts, and the connectors labeled A to D link one part to the next.

Part 1 represents the start-up. Upon power-up, the panel controller (ESP32) initializes its Wi-Fi, NTP, LoRa module, and PZEM-004T, then loads the stored schedule from its flash memory. The contactor is set to match the schedule (ON if the lamps should be lit at that moment, OFF otherwise), after which the controller connects to the Wi-Fi and the MQTT broker. Since the schedule is stored in the controller itself, the feed keeps switching on time even when the Wi-Fi or the server is down. The flow then proceeds to connector A.

Part 2 represents the switching control. The controller reads the time and checks whether a valid command from the dashboard is present. If there is one, the ON/OFF override is applied until it expires; otherwise, the controller follows the schedule (ON at 6:00 PM and OFF at 5:30 AM). The result drives the relay module, which switches the contactor, hence the entire feed is switched at once. The flow then proceeds to connector B.

Part 3 represents the polling process. The controller first reads the voltage, current, and power from the PZEM-004T. Since all lamp nodes share a single radio channel, the controller then polls them one at a time through LoRa instead of letting them transmit freely, which prevents the nodes from talking over each other. The addressed node wakes up, reads its light level

and battery level, and replies. If no reply arrives, the node is flagged as offline and the controller moves on to the next node. Once all nodes are polled, the readings (V, I, P, lux, and battery) are published to the MQTT broker. Connector A returns the controller to Part 2 for the next cycle, whilst connector C passes the readings to the server. [Polling interval and node wake time to be finalized.] [Behavior during a Wi-Fi outage, such as whether readings are buffered, to be decided.]

Part 4 represents the fault logic on the server. The MQTT broker receives the readings and the server stores them in the database. The server then evaluates them in order: absence of feed voltage is a mains or feed loss; a current drop of about one lamp (0.7 to 0.95 A) means one lamp is out, with the light readings of the nodes identifying which lamp; a node that did not reply is marked offline; and a low battery raises an alert for the battery to be swapped. The current-drop and lamp checks are only evaluated whilst the contactor is ON. All results are logged and the dashboard screens are updated.

Part 5 represents the dashboard access and commands. After a role-based login, officials only get read-only screens, whilst the Engineering Department can send an ON/OFF command or edit the schedule. The server logs each command together with its expiry and publishes it to the MQTT broker. Connector D passes the command to Part 2, where the panel controller applies it.

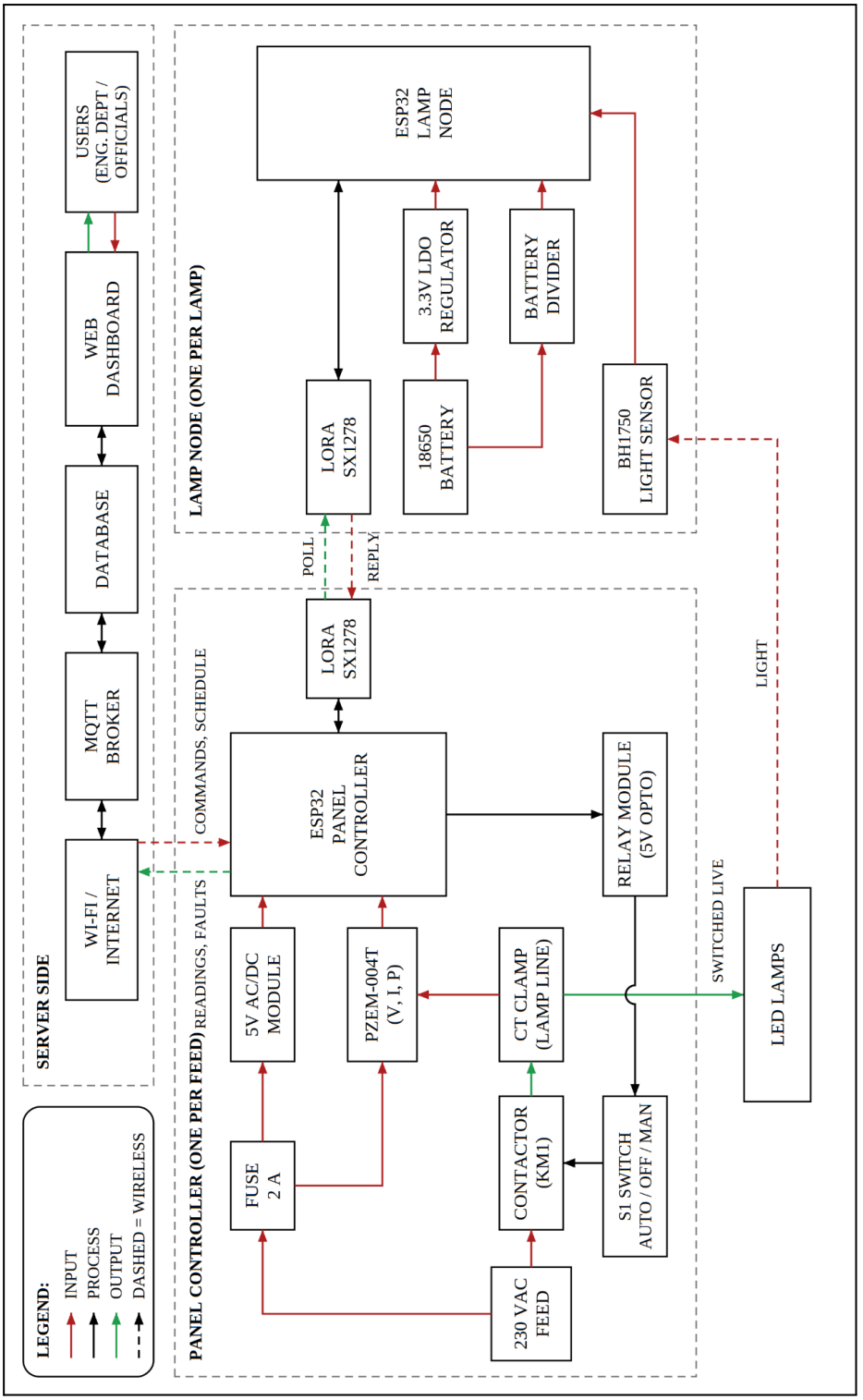


Figure 11. Functional Block Diagram

Functional block diagram. The functional block diagram in Figure 11 shows the overall functionality of the system by highlighting how power and data flow from the streetlights to the dashboard. It is divided into three subsystems: the lamp node, the panel controller, and the server side. As shown in the legend of the figure, red arrows represent inputs, black arrows represent processes, and green arrows represent outputs, whilst dashed arrows mark wireless links (LoRa, Wi-Fi, and light).

In the lamp node, the BH1750 receives the light of the lamp (input), the ESP32 wakes up upon the panel controller's poll and reads the light and battery levels (process), and the LoRa module sends the reading back (output). The node is powered by a replaceable 18650 battery through a 3.3 V regulator, whilst a battery divider lets the ESP32 read the remaining charge. The node has no charger; when the battery runs low, maintenance simply replaces it with a fresh one.

In the panel controller, the 230 VAC feed passes through a fuse to the 5 V AC/DC module, which powers the ESP32, and to the PZEM-004T, which measures the voltage of the feed. The CT clamp on the lamp line gives the PZEM-004T its current reading, and the PZEM-004T passes the measurements to the ESP32. The other inputs are the replies of the lamp nodes (through LoRa) and the schedule or dashboard commands (through Wi-Fi). The ESP32 handles the polling and switching logic (process), and its outputs are the relay module, the S1 selector, and the contactor that switch the lamp feed, together with the readings sent over Wi-Fi. The S1 selector allows the feed to be controlled manually if the ESP32 or the relay fails.

In the server side, the Wi-Fi and internet link carries the data to the MQTT broker, the database stores it, whilst the web dashboard displays the lamp statuses and fault history to the users (Engineering Department and officials). Commands sent from the dashboard travel back to the panel controller through the same path.

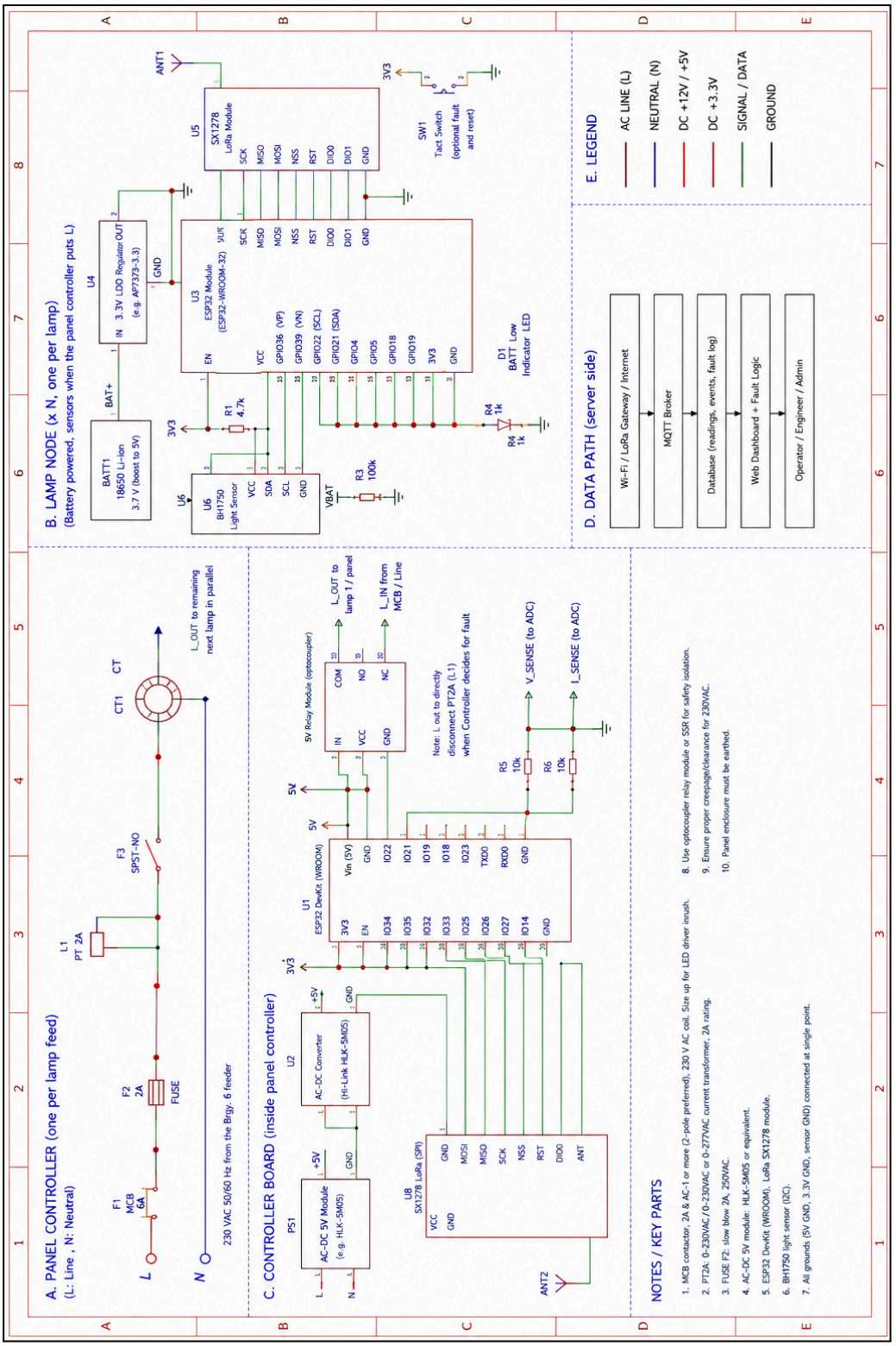


Figure 12. General Schematic Diagram

General schematic diagram. The general schematic diagram in Figure 12 shows the complete wiring of the panel controller and the lamp node. It illustrates the powering of each element and their interconnections, serving as the reference for building, testing, and validating the circuits whilst avoiding wiring errors.

In the panel controller, the 230 V feed passes through the existing MCB and a 2 A fuse (F1), which protects the control branch. The fused line supplies the 5 V AC/DC module, the voltage input of the PZEM-004T, and the coil of the contactor (KM1). The AC/DC module powers the ESP32, the relay module, and the SX1278 through its 5 V and 3.3 V rails. The PZEM-004T measures the current through a split-core CT clamped on the live wire of the lamp line only, and it communicates with the ESP32 through UART (9600 baud, 8N1). Since the PZEM-004T transmits at 5 V logic, a 1 kΩ/2 kΩ divider reduces the signal before it enters the ESP32. The SX1278 communicates through SPI and is powered from the 3.3 V rail since the module is not designed for 5 V. The ESP32's output pin drives the opto-isolated relay module, which in turn energizes the coil of the contactor through the S1 selector. When S1 is set to MAN, the coil is fed directly, which bypasses the ESP32 and the relay. For safety, the mains-side components are wired by the Engineering Department or a licensed electrician inside a closed enclosure.

In the lamp node, a protected 18650 battery in its holder supplies a low-quiescent-current 3.3 V regulator, which powers the ESP32 module, the SX1278, and the BH1750. The SX1278 communicates with the ESP32 through SPI, whilst the BH1750 communicates through I2C (SDA and SCL

lines). Two 100 kΩ resistors form a divider that lets the ESP32 read the battery voltage on an ADC pin. The node has no connection to the lamp's mains and only watches its light.

3D model of the prototype. The 3D models visually represent the structure, layout, and intended functionality of the two hardware units before actual physical construction. Each model is presented in isometric, front (with dimensions), rear, right, left, top, and bottom views.

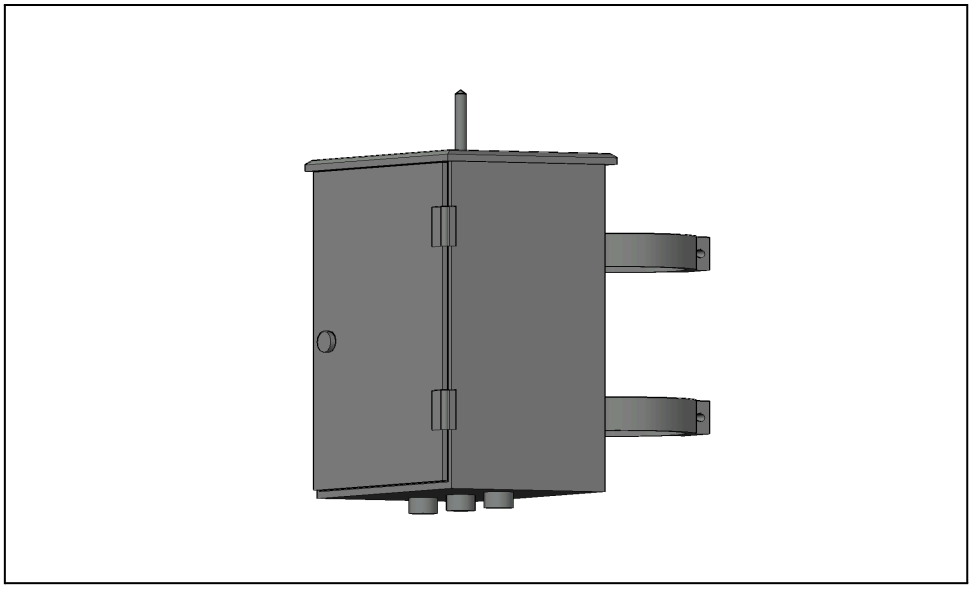


Figure 13. *Isometric view of the prototype*

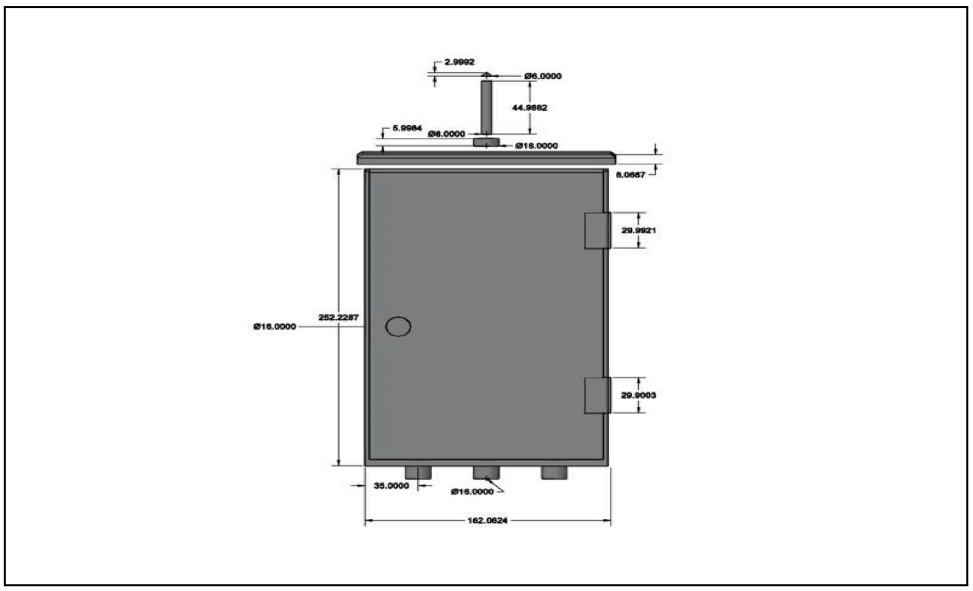


Figure 14. *Front view with dimensions (162.06 mm width × 252.23 mm height)*

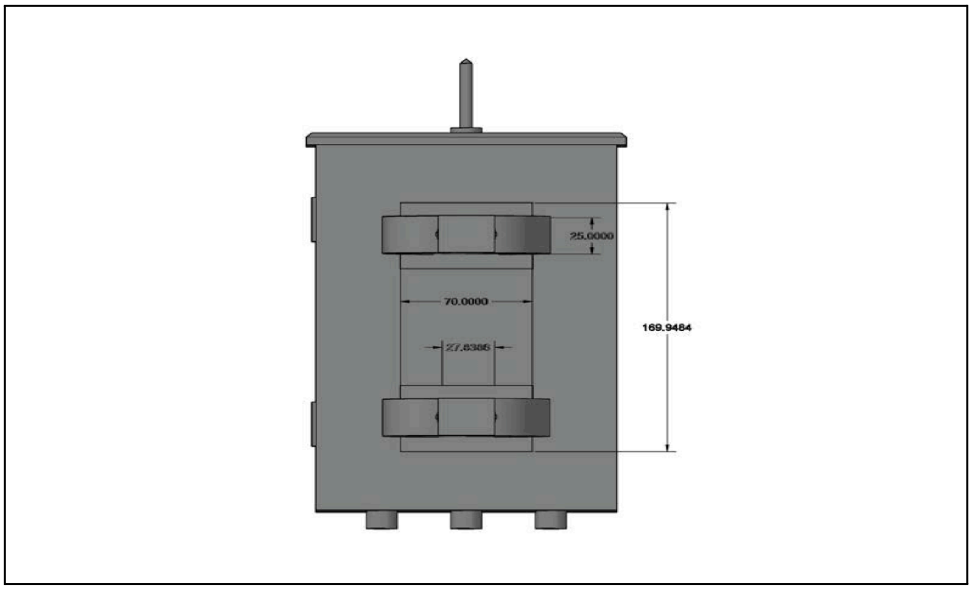


Figure 15. *Rear view with dimensions (169.95 mm height span)*

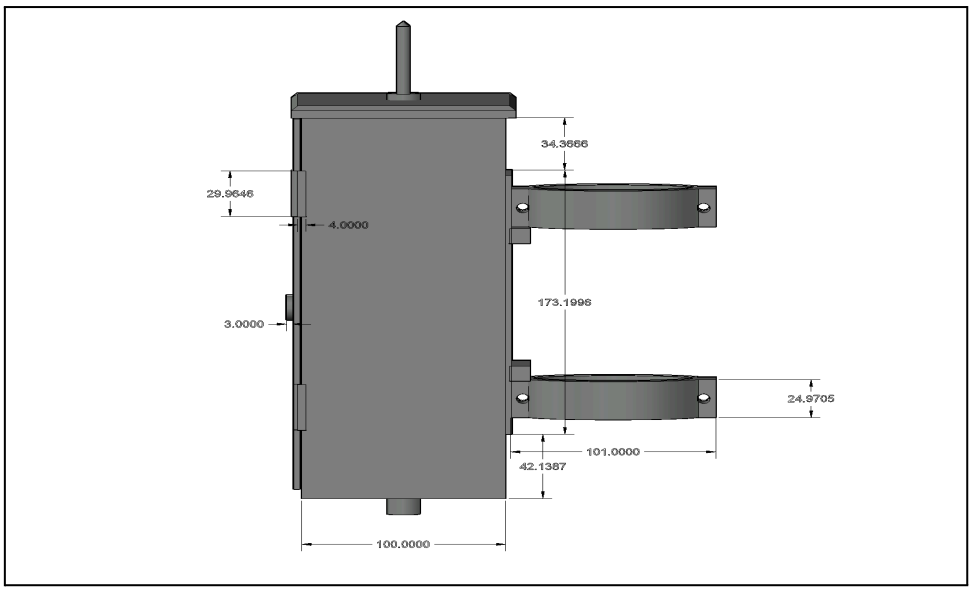


Figure 16. *Right view with dimensions (100.00 mm depth, 173.20 mm mount height)*

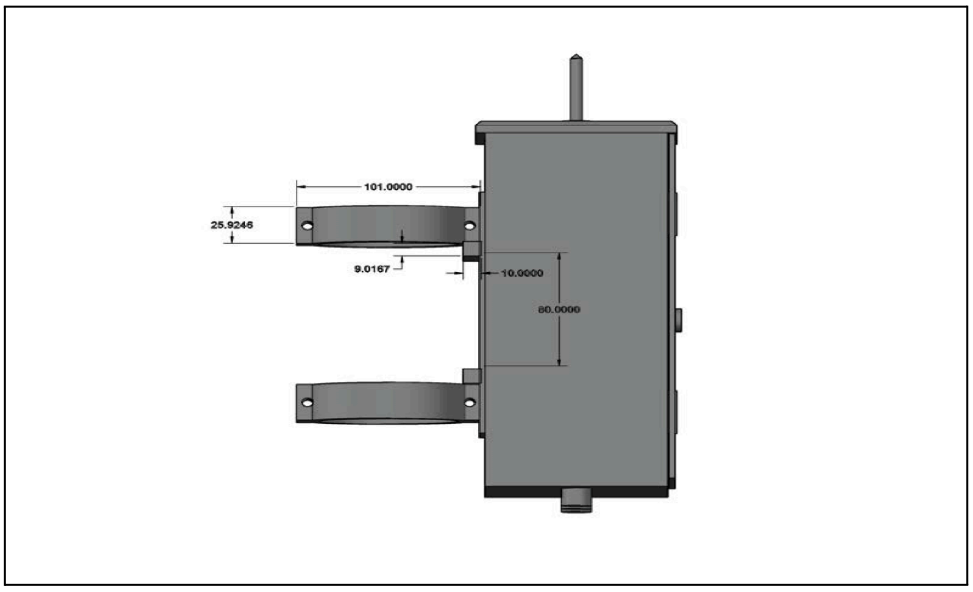


Figure 17. *Left view with dimensions (101.00 mm bracket depth, 80.00 mm mounting center)*

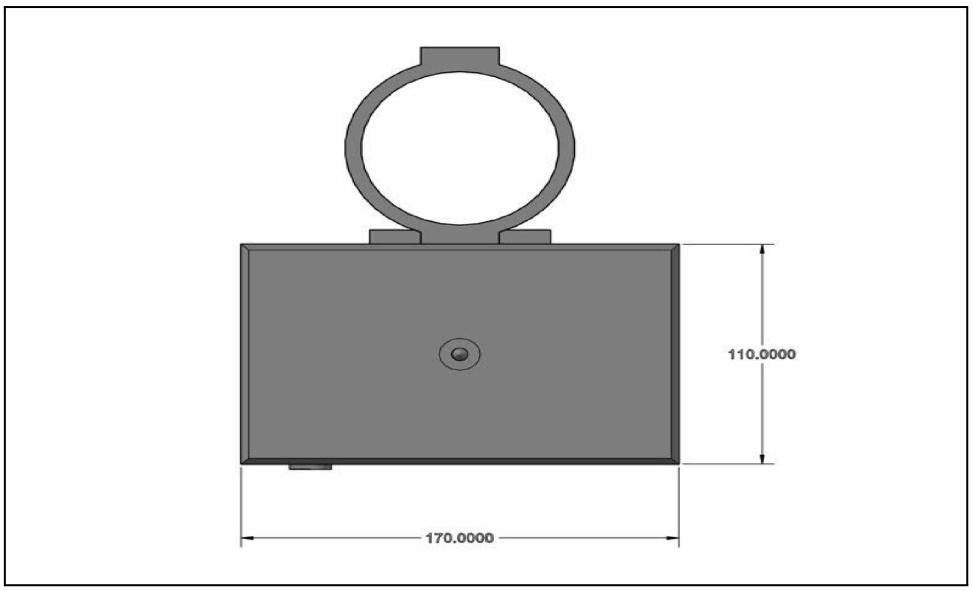


Figure 18. *Top view with dimensions (170.00 mm width × 110.00 mm depth)*

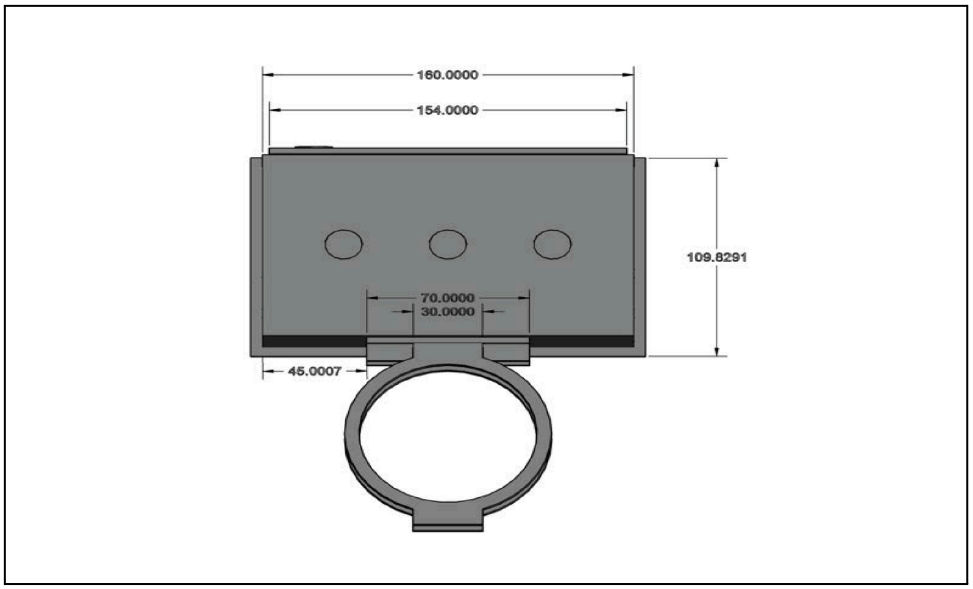


Figure 19. *Bottom view with dimensions (160.00 mm base width × 109.83 mm depth)*

Figures 13 to 19 show the 3D models and precise dimensions of the hardware units. The panel controller enclosure measures 162.06 mm in front

width, 252.23 mm in main body height (excluding the top antenna mount), and 100.00 mm in depth, with a mounting bracket depth of 101.00 mm. The compact lamp node housing measures 170.00 mm in top width by 110.00 mm in top depth, with a base enclosure dimension of 160.00 mm by 109.83 mm.

The panel controller is a weatherproof box mounted at the power feed, with the LoRa antenna on top, cable glands at the bottom for the feed and CT wires, and the manual override switch accessible on the front. Internal mounting plates hold the power supply, ESP32, relay module, contactor, and terminal blocks, built within the 252.23 mm × 162.06 mm frame.

Parts of the prototype. The key components of the prototype enable the system to sense the lamps, communicate wirelessly, switch the feed, and report to the dashboard. Each component has a designated place in its unit's housing, ensuring that the system runs efficiently during operation.

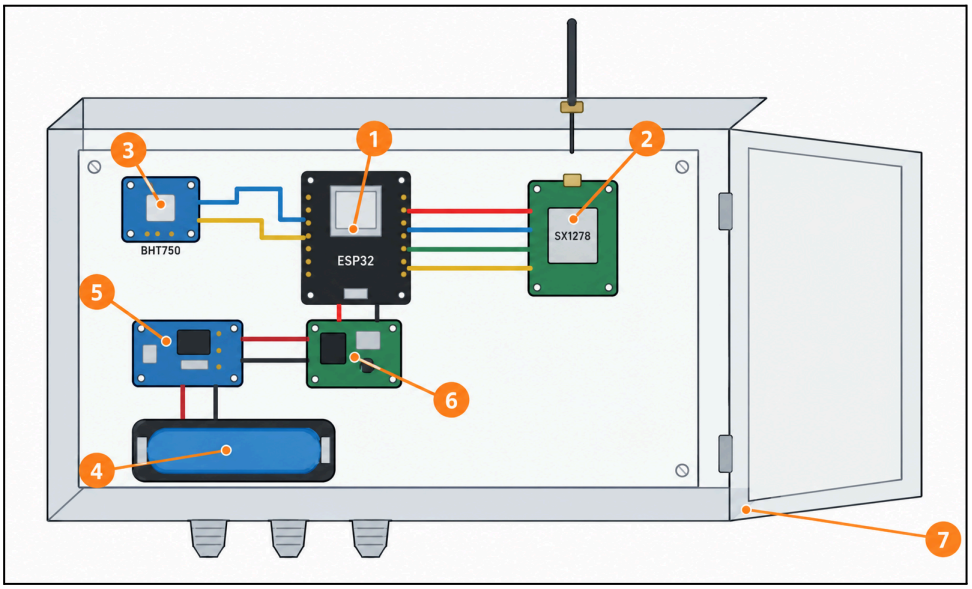


Figure 20. Parts of the Prototype (Lamp Node Mockup)

In the lamp node (Figure 20), the ESP32-WROOM-32 [1] serves as the controller, reading the light sensor and answering the panel controller's requests. The SX1276/SX1278 LoRa module and antenna [2] handle the wireless communication, whilst the BH1750 [3] measures the light of the lamp. The lithium-ion battery [4] supplies the power, charged through the TP4056 module [5] and stepped up by the boost converter [6]. All of these are enclosed in the project box [7].

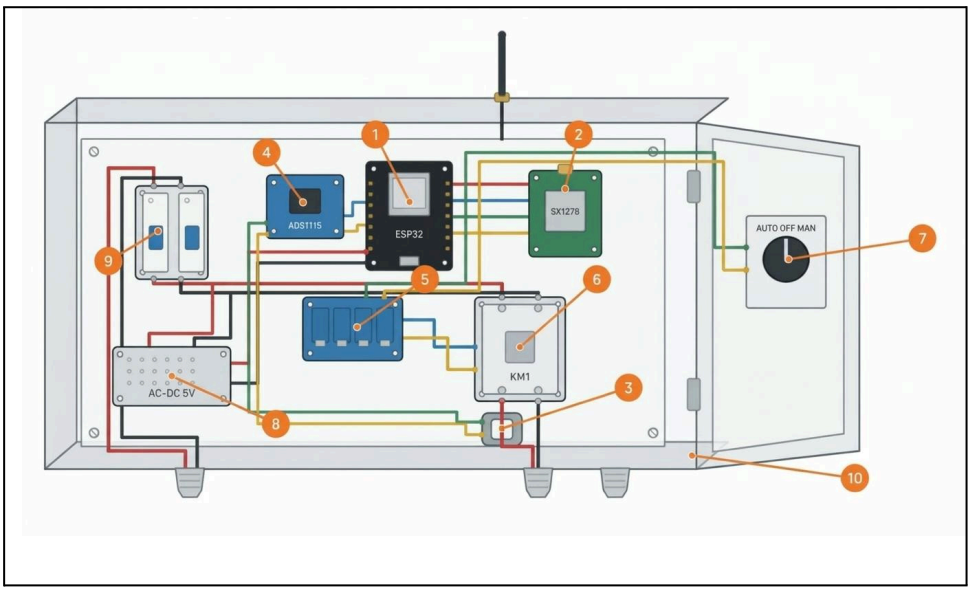


Figure 21. Parts of the Prototype (Circuit/Breaker Panel Mockup)

In the panel controller (Figure 21), the ESP32-WROOM-32 [1] serves as the main controller, communicating with the lamp nodes through the SX1276/SX1278 LoRa module and antenna [2] and with the server over Wi-Fi. The CT sensor [3] and ADS1115 [4] measure the feed's current, whilst the relay module [5] and contactor [6] switch the feed, with the manual override switch [7] as the physical backup. The AC-DC power supply [8]

powers the electronics, the circuit protection set [9] guards them, and the weatherproof enclosure [10] houses everything.

Software development. The software development component outlines the approach used in programming and integrating the firmware, server, and dashboard needed for centralized switching and per-lamp fault monitoring. This section describes how the firmware of the two hardware units, the data pipeline, and the web dashboard are developed to ensure that each software layer works together in support of reliable switching, fault detection, and monitoring.

Materials needed for the software. The following software tools are used during the development process, each selected for its compatibility with the ESP32 and its availability at no cost.

Arduino IDE. The Arduino IDE is an open-source application used to write, compile, and upload the C++ firmware of both ESP32 boards. It also provides the libraries for the LoRa module and the BH1750 sensor.

C++ (Arduino framework). C++ is the programming language of the firmware, used for the sensor reading, LoRa polling, current computation, and contactor control.

Visual Studio Code. Visual Studio Code serves as the development environment for the server and dashboard code.

MQTT broker (Eclipse Mosquitto). MQTT is a lightweight messaging protocol suited to small devices. The broker receives the readings

published by the panel controller and passes them to the server, hence the controller does not need to talk to the database directly.

MySQL. MySQL is the database that stores the lamp records, readings, fault history, and switching events. [Database system to be confirmed.]

Web dashboard stack. The dashboard is built using [HTML, CSS, and JavaScript, with a backend language to be confirmed] so that it can be opened from any browser without installing an application.

Software walkthrough. The web dashboard is designed for the monitoring and control of the streetlights. It serves two user groups:

(1) Engineering Department personnel, who view the lamp statuses and fault history and send switching commands; and

(2) Barangay 6 officials, who view the lamp statuses and operating records.

Login page. As shown in Figure 22, the login page is the system's entry point, requiring a username and password so that only authorized users can view the lamps or send commands.

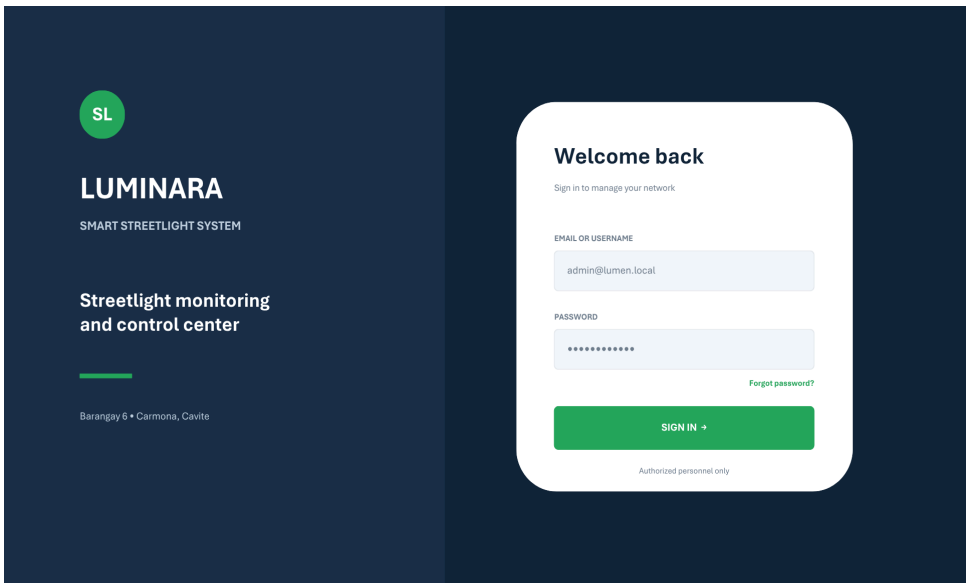


Figure 22. *Login page interface of the web dashboard*

Dashboard overview. Figure 23 shows the overview page, which presents the total number of lamps, how many are ON, OFF, or faulty, and the operating hours of the feed. Below it, a lamp grid (or map) shows each lamp's location and live status.

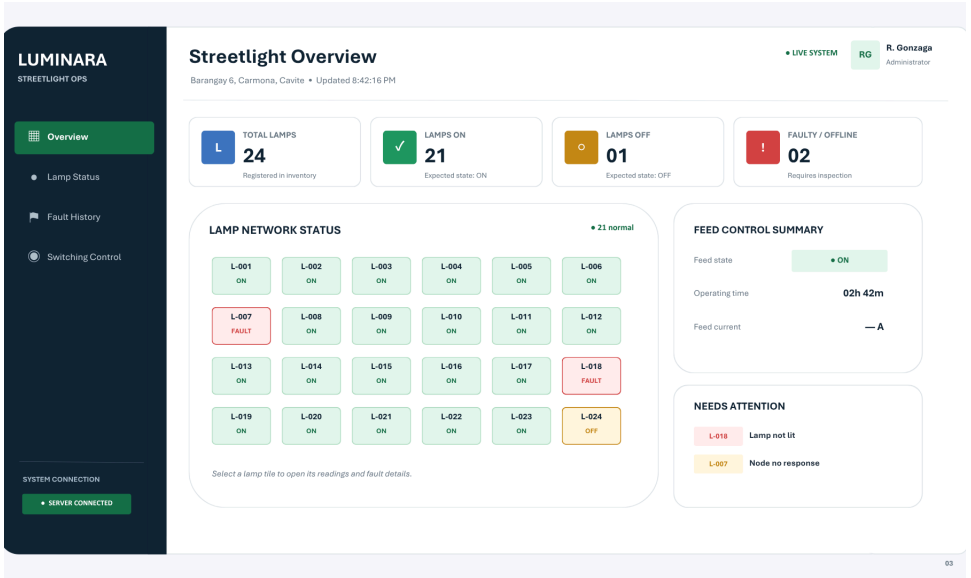


Figure 23. *Dashboard overview interface*

Lamp status and fault alerts. As illustrated in Figure 24, a lamp with a fault is highlighted together with its fault class (lamp not lit, unstable, or no response) and the latest light reading, so that the exact lamp to be repaired can be identified at a glance.

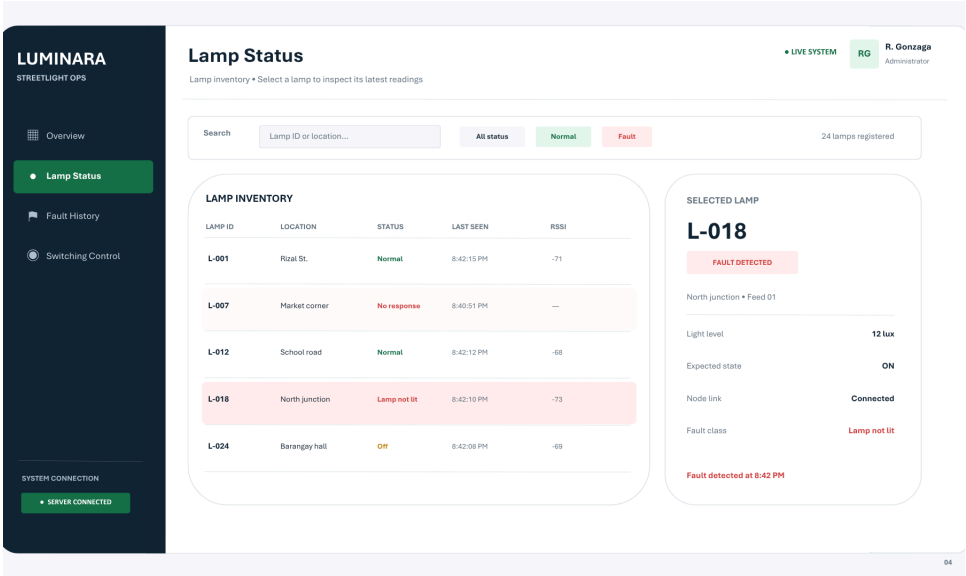


Figure 24. *Lamp status and fault alerts view*

Fault history. Figure 25 shows the fault history, a searchable table of past faults with the lamp ID, fault class, time detected, and time resolved.

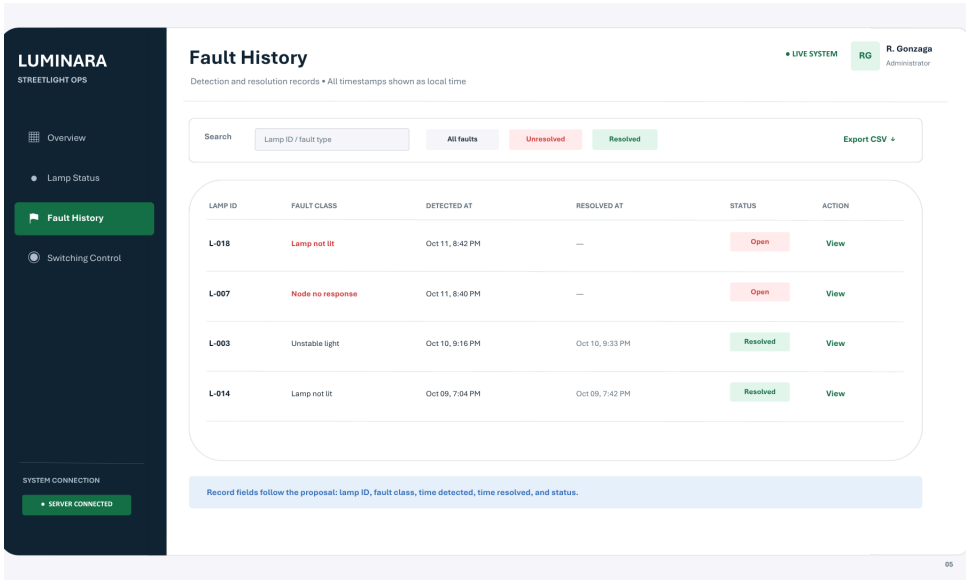


Figure 25. *Fault history interface*

Switching control. Figure 26 shows the switching control page, where authorized users can send ON/OFF commands to the feed and view the schedule and the log of switching events.

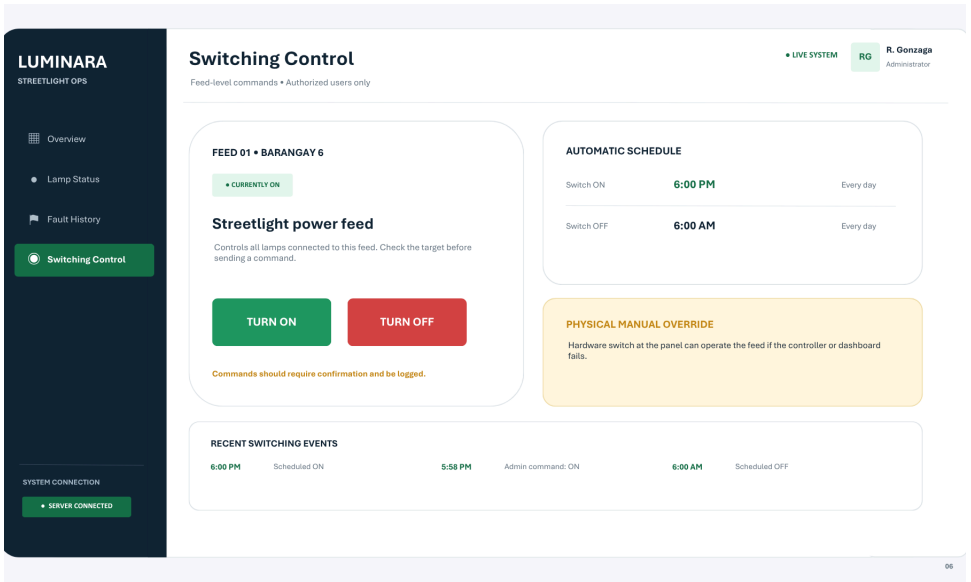


Figure 26. *Switching control interface*

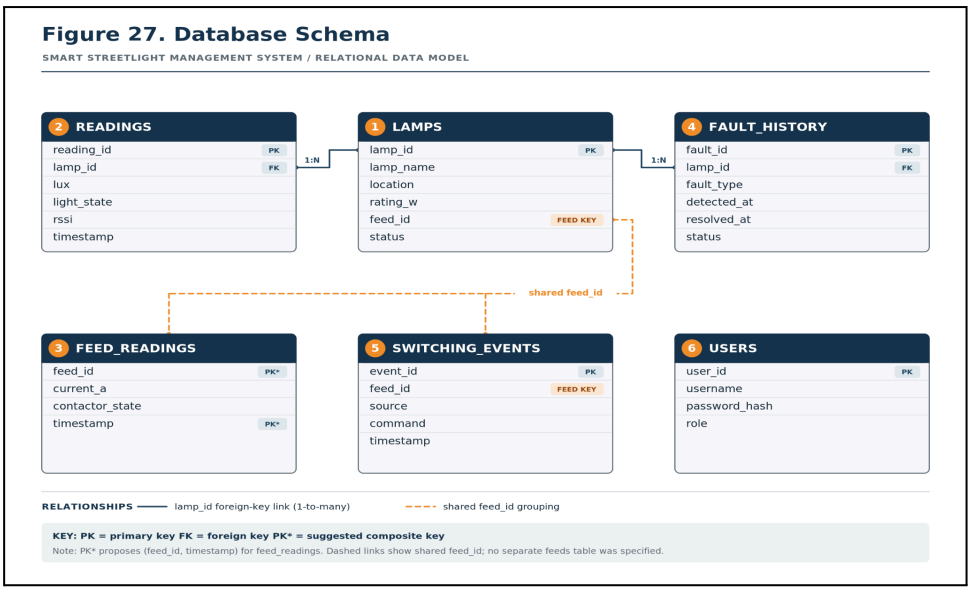


Figure 27. Database Schema

Database design. Figure 27 shows the database schema. It consists of the following tables: (1) lamps (lamp_id, lamp_name, location, rating_w, feed_id, status); (2) readings (reading_id, lamp_id, lux, light_state, rssi, timestamp); (3) feed_readings (feed_id, current_a, contactor_state, timestamp); (4) fault_history (fault_id, lamp_id, fault_type, detected_at, resolved_at, status); (5) switching_events (event_id, feed_id, source, command, timestamp); and (6) users (user_id, username, password_hash, role).

Methods

The methodology of this study presents the development approach and the procedures followed in the development, testing, data collection, and evaluation of the Smart Streetlight Management System.

The development of the system follows the Prototyping model of the Software Development Life Cycle (SDLC), applied incrementally. In this model, a working prototype is built and tested early so that the actual behavior of the system can be observed and corrected, and the prototype is then grown one function at a time instead of being built whole. Each increment is tested, reviewed with the adviser, and refined before the next one is added. Prototyping was selected because the system integrates hardware and software whose real behavior (the LoRa range between poles, the readings of the light sensor at the lamp, and the current readings on the feed) can only be seen once the parts are actually built and tested (Maryani et. al., 2022). Building the prototype in increments follows the long-established practice of iterative and incremental development, which avoids a single-pass, sequential, document-driven process (Larman & Basili, 2003). It also suits a four-member group working under a limited budget, since a problem found in one increment does not require rebuilding the entire system (Alshamrani & Bahattab, 2015).

PHASE	SDLC STAGE	STREETLIGHT ACTIVITIES
1	Requirements Analysis and Documentation	Interview the Engineering Department and Barangay 6 officials; conduct a site survey of the lamps (count, wiring, ratings, and spacing); define the fault classes; and select the hardware within the budget.
2	System Design	Develop the functional block diagram, general schematic diagrams, and 3D models; design the MySQL database schema; and create the dashboard wireframes.
3	Prototype Development and Testing (by Increment)	**Increment 1**: panel controller with scheduled contactor switching, manual override, and CT current reading. **Increment 2**: lamp node with BH1750 reading, LoRa reply, and battery power. **Increment 3**: polling of multiple lamp nodes and fault classification. **Increment 4**: MQTT, database, and web dashboard. Each increment is tested, reviewed with the client, and refined before the next.
4	System Testing	Conduct integration testing and system testing on the complete prototype.
5	Evaluation	Administer the ISO/IEC

		25010 evaluation instrument to Brgy. 6 respondents; compute the weighted mean per quality characteristic; and interpret the results against the mean range scale.
6	Implementation Plan	Prepare the plan for potential pilot adoption by the Carmona Engineering Department.

Table 2. Prototyping phases of the system, with the prototype built in increments.

Data collection. The data collected in this study come from three sources: the interviews and site survey before development, the readings of the prototype during testing, and the responses of the evaluators. Table 3 details the data the prototype records.

DATA	SOURCE	COLLECTED THROUGH	PURPOSE
Site survey and interview records (once, before development)	Engineering Department and Barangay 6 officials	Interview and physical survey	Determine lamp count, wiring, and fault concerns
Light reading (lux)	BH1750 at each lamp node	Read by the lamp node and sent via LoRa when polled	Confirm if each lamp is lit; detect flickering
Feed current	CT sensor	RMS	Confirm the feed

		computation by the ESP32	is under load
LoRa link status	Panel controller's LoRa module	Poll reply or timeout; RSSI read per received packet	Detect unreachable nodes; assess link reliability
Fault events (upon detection)	Fault classification rules	Logged automatically	Fault history and alerts
Switching events (upon execution)	Panel controller	Logged automatically	Record operating hours and commands

Table 3. *Data collection matrix*

Testing. The prototype will undergo unit testing (each component in isolation), integration testing (the combined subsystems), and system testing (the complete system under simulated conditions). Table 4 shows the test cases. The Actual Result and Status columns will be added once testing begins.

TEST ID	CATEGORY	TEST	PROCEDURE	EXPECTED RESULT
UT-01	Unit	CT sensor accuracy	Apply a known AC load	Reading within ±5% of a reference clamp meter [tolerance to be confirmed]
UT-02	Unit	BH1750 response	Switch the lamp on and off in front of	Lit and unlit readings clearly

			the sensor	separated, compared against a reference lux meter
UT-03	Unit	Contactor switching	Send a GPIO command	Contactor closes and opens, and the lamps switch accordingly
UT-04	Unit	Node battery runtime	Run a lamp node on battery under the polling schedule	Node operates for [target duration]
IT-01	Integration	Node-to-panel link	Poll all lamp nodes in sequence	Each node replies; packet success rate and RSSI recorded against [target]
IT-02	Integration	Panel-to-dashboard	Publish readings via MQTT	Entries appear in the database and the dashboard shows the correct lamp status
IT-03	Integration	Dashboard command	Click ON/OFF on the dashboard	Contactor switches within 2-4 seconds
ST-01	System	Fault detection	Induce each fault class on a mock lamp setup: lamp off, flickering, node off, and feed breaker open	Each fault is flagged within 1-3 polling cycles, with the lamp ID for lamp faults and a

				single feed-level flag for the breaker; Fault Detection Rate recorded
ST-02	System	Manual override	Turn the override ON during an OFF command	Lamps turn ON regardless of the controller
ST-03	System	Scheduled cycle	Run the ON/OFF schedule for [24 hours]	Switching on time with complete logs

Table 4. Testing Matrix

Evaluation and validation. The completed prototype and dashboard will be evaluated using an instrument adapted from the ISO/IEC 25010 software product quality model (International Organization for Standardization & International Electrotechnical Commission [ISO/IEC], 2011). The evaluators will consist of [20] respondents composed of IT/Engineering experts, Barangay 6 officials, and Carmona Engineering Department personnel. The system will be rated on the following characteristics:

1. **Functional suitability:** how complete and correct the switching and fault detection functions are;
2. **Performance efficiency:** the response time of switching and fault detection;

3. **Compatibility:** how well the lamp nodes, panel controller, server, and dashboard work together;
4. **Usability:** the ease of using and understanding the dashboard;
5. **Reliability:** how consistently the system operates, including the delivery of the lamp nodes' data;
6. **Security:** the protection of the dashboard through user login;
7. **Maintainability:** how easily parts (such as batteries and nodes) can be replaced or the firmware updated; and
8. **Portability:** how easily the system can be installed on other feeds or streets.

A 5-point Likert scale (5 - Strongly Agree, 4 - Agree, 3 - Neutral, 2 - Disagree, 1 - Strongly Disagree) will be utilized for evaluation scoring.

SCALE	DESCRIPTION
5	Strongly Agree
4	Agree
3	Neutral
2	Disagree
1	Strongly Disagree

Table 5. Likert Scale

Data Analysis. The data from testing and evaluation will be analyzed using the following measures:

Fault Detection Rate (FDR) = (Faults Detected ÷ Faults Induced) × 100

Switching Success Rate (SSR) = (Successful Switching Operations ÷ Total Switching Attempts) × 100

Packet Success Rate (PSR) = (Packets Received ÷ Packets Sent) × 100

Average Response Time (ART) = Σ response times ÷ number of trials

Weighted Mean ($\bar{X}$) = $\Sigma(f \cdot x) \div N$

where f is the frequency of responses for a rating, x is the rating's weight on the Likert scale, and N is the total number of evaluators.

Implementation plan. The following plan outlines the steps for the potential pilot adoption of the system by the Carmona Engineering Department in Barangay 6.

Phase 1: *Site survey and asset inventory.* Record the pole locations, feed configuration, lamp ratings, and spacing, and check the LoRa line of sight between poles.

Phase 2: *Procurement and assembly.* Source the components and assemble the panel controller and lamp nodes.

Phase 3: *Installation.* Install the panel controller at the feed's switchboard with the CT sensor clamped around the live line, and mount a lamp node beside each lamp with its BH1750 aimed at the bulb.

Phase 4: *Server and network configuration.* Set up the MQTT broker, database, and dashboard, and register each lamp's ID and location.

Phase 5: *Calibration and commissioning.* Set the light threshold per lamp and the baseline feed current, then test the manual override.

Phase 6: *Training and handover.* Train Engineering Department and barangay personnel on the dashboard, battery replacement, and basic troubleshooting.

Phase 7: *Pilot monitoring.* Monitor the system over an initial [30-day] window, logging faults and the repair response.

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